

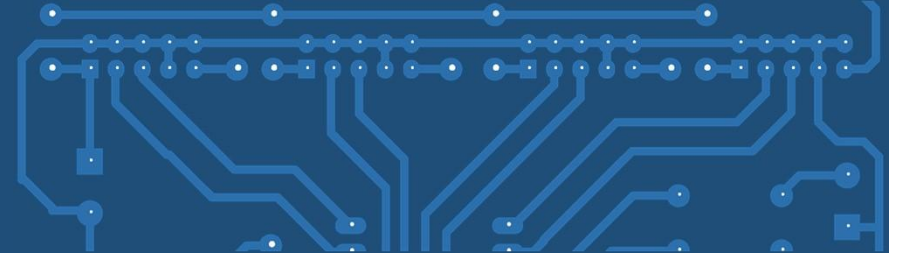
Topic 2 Reference Frames and Orientation

Part 1

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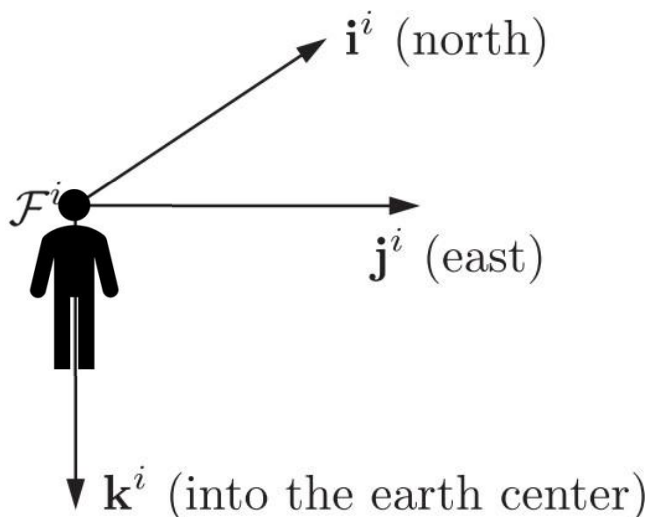
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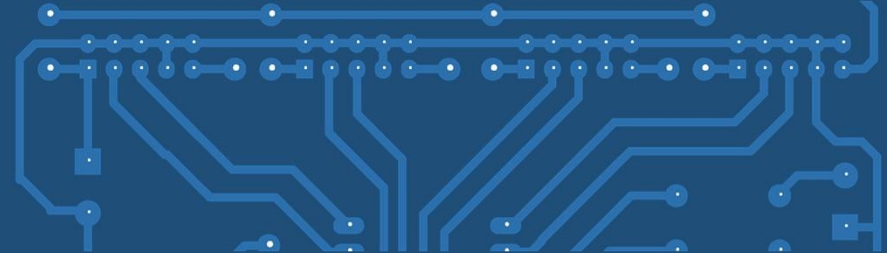
Obvious reference frames are:

- Body reference frame (local)
- Inertial reference frame (global)
Why is it called inertial?



Are these reference frames stationary?

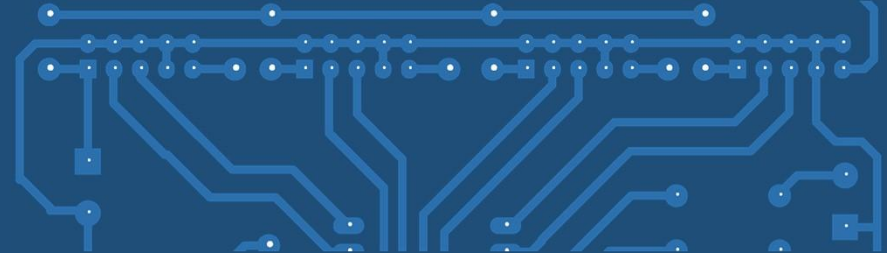
- Inertial reference frame
Yes, fixed to the environment
- Body reference frame
No, fixed to the object → moving, and changing orientation



Why different coordinate frames?

- Newton's equations of motion are derived relative to:
a fixed inertial reference frame.
- Motion is most easily described in:
a body-fixed frame.
- Aerodynamic forces and torques act on the aircraft body and are most easily described in:
a body-fixed reference frame.
- On-board sensors like accelerometers and rate gyros measure information with respect to:
the body frame.
- GPS measures position, ground speed, and course angle with respect to:
the inertial frame.
- Mission requirements, like loiter points and flight trajectories, are specified in:
the inertial frame.
- Map information is also given in:
an inertial frame.





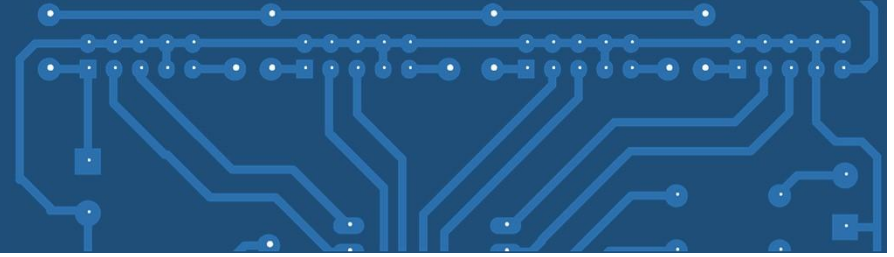
Aerial robotics coordinate frames:

- the inertial frame,
- the vehicle frame,
- the vehicle-1 frame, the vehicle-2 frame, the body frame,
- the stability frame, and the wind frame.

Notes:

- The inertial and vehicle frames are related by a **translation**, while the remaining frames are related by **rotations**.
- The angles defining the relative orientations of the vehicle, vehicle1, vehicle-2, and body frames are the **roll**, **pitch**, and **yaw** angles that describe the attitude of the aircraft. These angles are commonly known as **Euler angles**.
- The rotation angles that define the relative orientation of the body, stability, and wind coordinate frames are the **angle-of-attack** and **side-slip angles**.
- Throughout the course we assume a flat, non-rotating earth.
- The aerial robot is assumed to be a rigid body.

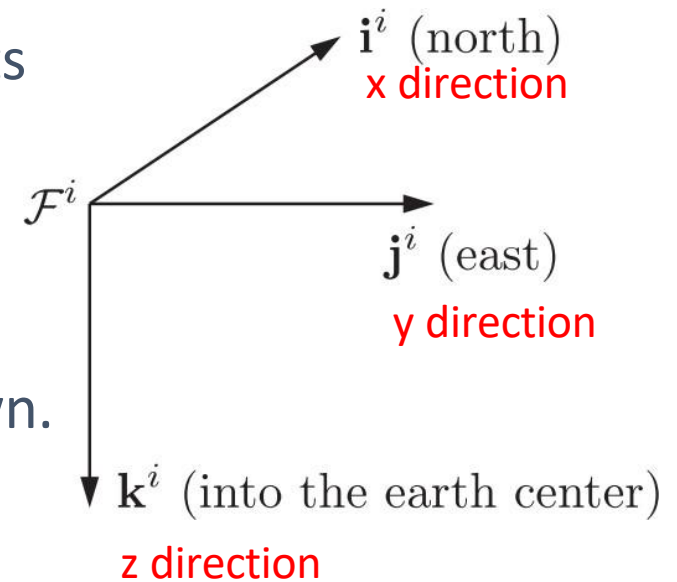




The inertial frame $\mathcal{F}^i$

Is an earth-fixed coordinate system with its origin at the defined home location.

Unit vector $\mathbf{i}^i$ is directed north,
 $\mathbf{j}^i$ is directed east, and
 $\mathbf{k}^i$ is directed toward the
center of the earth, or down.



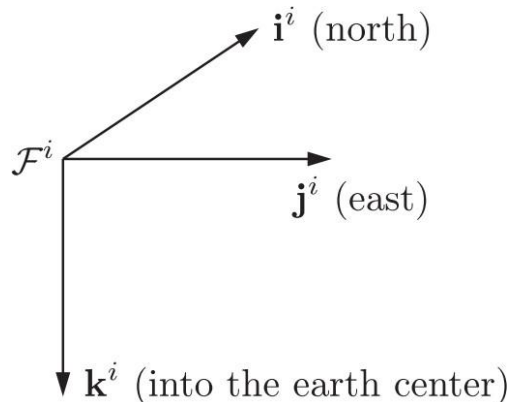
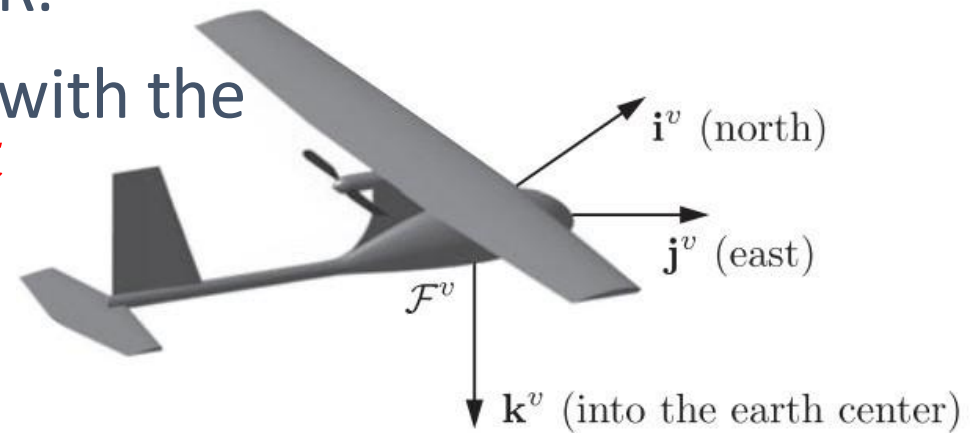
Referred to as a North-East-Down (NED)
reference frame.

It is common for **north** to be referred to as the inertial **x direction**,
east to be referred to as the inertial **y direction**, and
down to be referred to as the inertial **z direction**.

The vehicle frame $\mathcal{F}^v$

The origin of the vehicle frame is at the center of mass of the AR.

The axes of $\mathcal{F}^v$ are aligned with the axis of the inertial frame $\mathcal{F}^i$

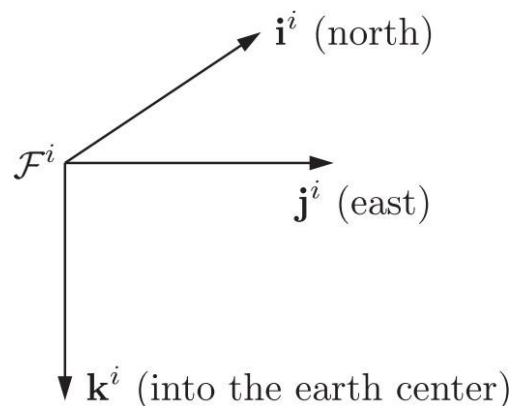
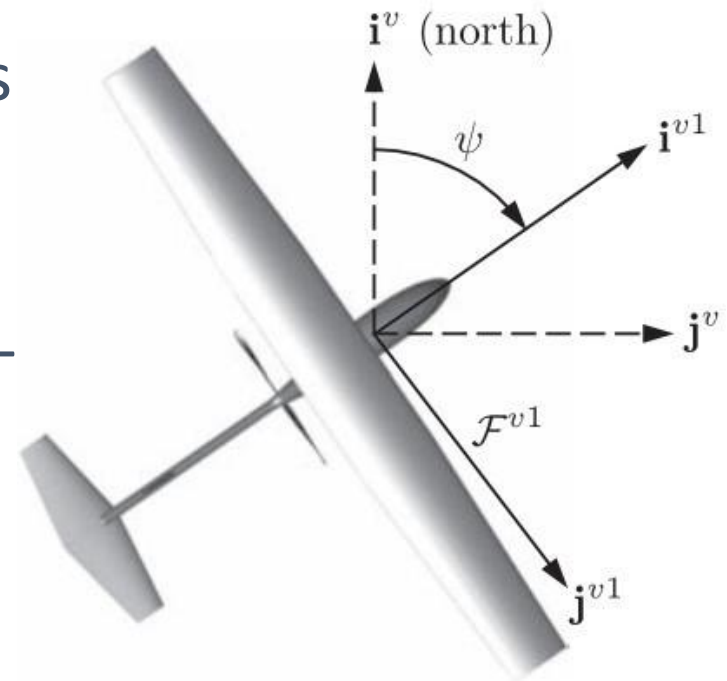


$\mathbf{i}^v$ —axis points to north
 $\mathbf{j}^v$ —axis points to east
 $\mathbf{k}^v$ —axis points into the earth

The vehicle-1 frame $\mathcal{F}^{v1}$

The origin of the vehicle-1 frame is identical to the vehicle frame, i.e. CoM of the AR.

$\mathcal{F}^{v1}$ is rotated in the positive right-handed direction about $\mathbf{k}^v$ by the heading (or **yaw**) angle Ψ .



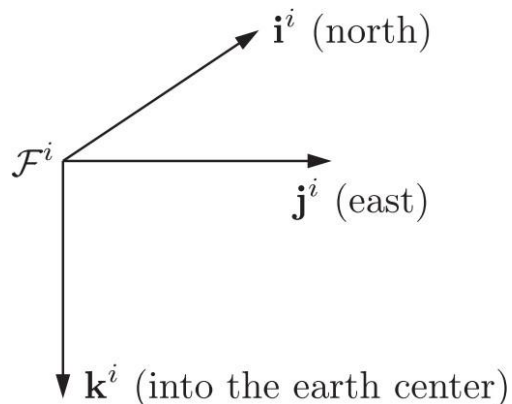
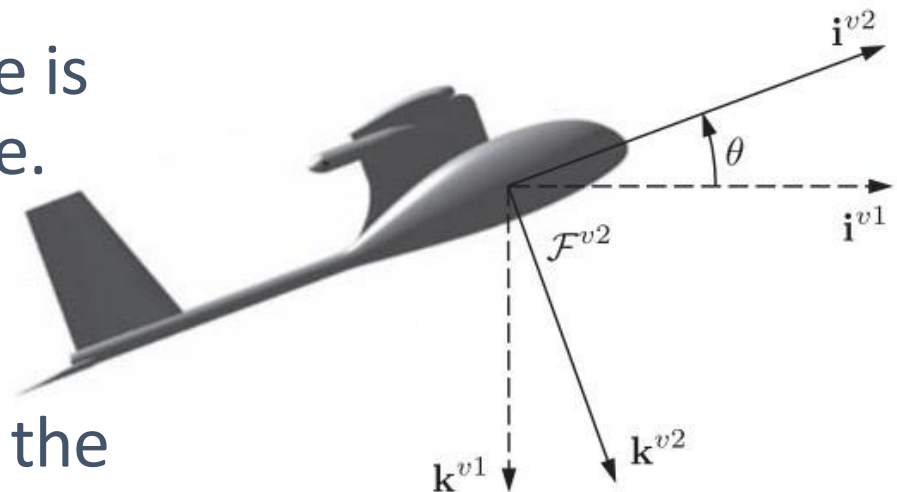
$\mathbf{i}^{v1}$ —axis points out the **nose** of the AR
 $\mathbf{j}^{v1}$ —axis points out the **right** of the AR
 $\mathbf{k}^{v1}$ —axis points into the **earth**.



The vehicle-2 frame $\mathcal{F}^{v2}$

The origin of the vehicle-2 frame is identical to the vehicle frame, i.e. CoM of the AR.

$\mathcal{F}^{v2}$ is obtained by rotating $\mathcal{F}^{v1}$ in a righthanded rotation about the $\mathbf{j}^{v1}$ axis by the **pitch** angle θ .

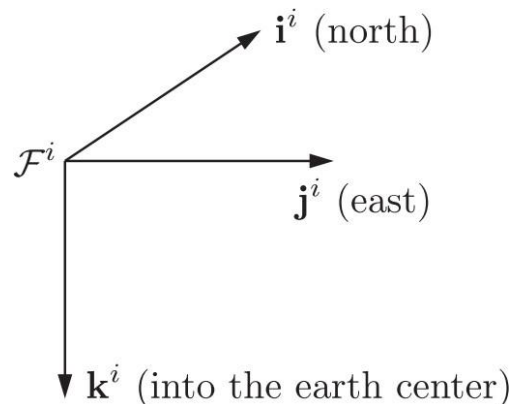
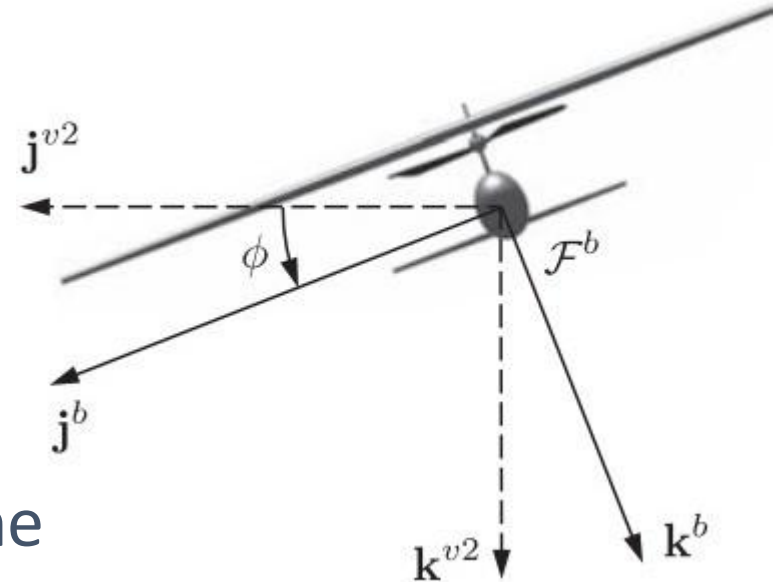


$\mathbf{i}^{v2}$ —axis points out the **nose** of the AR
 $\mathbf{j}^{v2}$ —axis points out the **right** of the AR
 $\mathbf{k}^{v2}$ —axis points out the **belly** of the AR

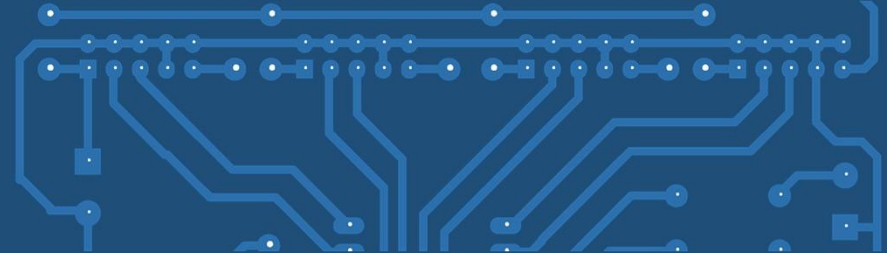
The body frame $\mathcal{F}^b$

The origin of the body frame is identical to the vehicle frame, i.e. CoM of the AR.

$\mathcal{F}^b$ is obtained by rotating $\mathcal{F}^{v2}$ in a righthanded rotation about the $\mathbf{i}^{v2}$ axis by the **roll** angle ϕ .



$\mathbf{i}^b$ –axis points out the **nose** of the AR
 $\mathbf{j}^b$ –axis points out the **right** of the AR
 $\mathbf{k}^b$ –axis points out the **belly** of the AR



Transformation from the vehicle frame $\mathcal{F}^v$ to the body frame $\mathcal{F}^b$

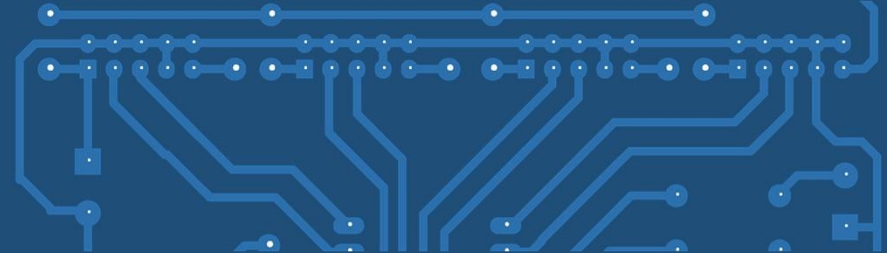
The origin of the two frames are identical, i.e. CoM of the AR.

The rotation sequence $\psi - \theta - \phi$ is commonly used for aircraft and is just one of several Euler angle systems in use.

The physical interpretation of Euler angles is clear and this contributes to their widespread use.

Euler angle representations, however, have a mathematical singularity that can cause computational instabilities. For the $\psi - \theta - \phi$ Euler angle sequence, there is a singularity when the pitch angle θ is ± 90 deg, in which case the yaw angle is not defined. This singularity is commonly referred to as **gimbal lock**.





Coordinate frames for fixed-wing ARs

Aerodynamic forces are generated as the airframe moves through the air surrounding it.

The velocity of the aircraft relative to the surrounding air is referred to as the airspeed vector V_a .

The magnitude of the airspeed vector is simply referred to as the airspeed, V_a .

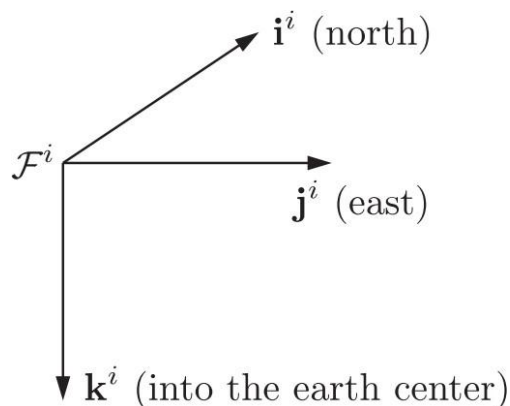
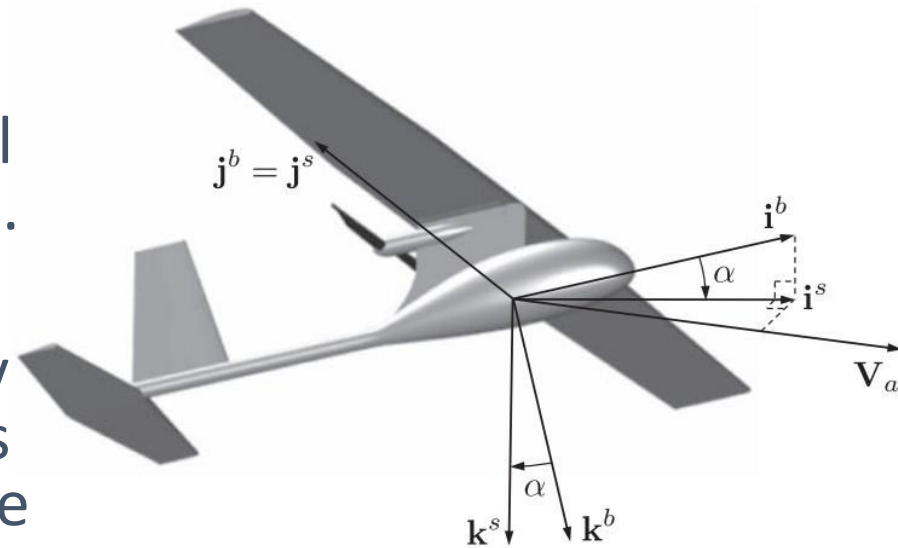
Angle of attack, α , describes the angle between the wing chord line and the incoming airflow.



The stability frame $\mathcal{F}^s$

The origin of the body frame is identical to the vehicle frame, i.e. CoM of the AR.

$\mathcal{F}^s$ is obtained by rotating $\mathcal{F}^b$ in a **lefthanded** rotation about the $\mathbf{j}^b$ axis by the **angle-of-attack** α , such that $\mathbf{i}^s$ aligns with the projection of V_a onto the plane spanned by $\mathbf{i}^b$ and $\mathbf{k}^b$.



$\mathbf{i}^s$ -axis points along the projection of the airspeed vector onto the $\mathbf{i}^b$ - $\mathbf{k}^b$ plane of the body frame

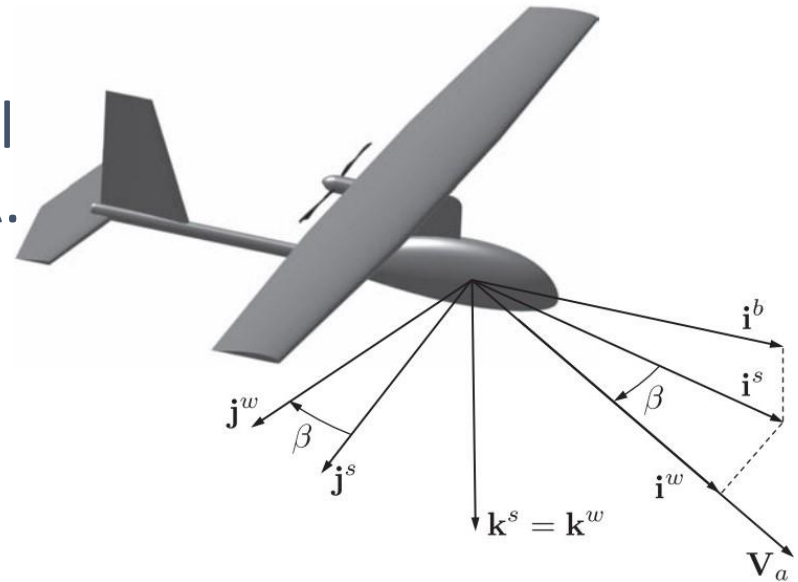
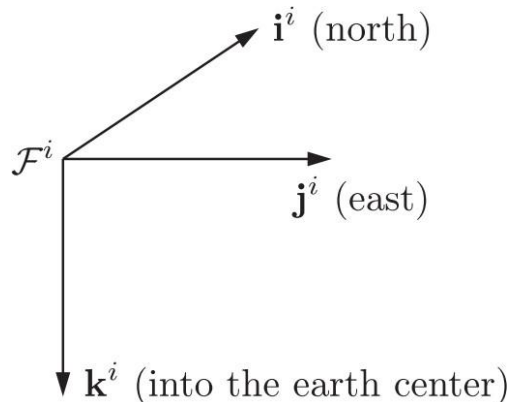
$\mathbf{j}^s$ -axis is identical to $\mathbf{j}^b$ -axis

$\mathbf{k}^s$ -axis is constructed to make a right-handed coordinate system.

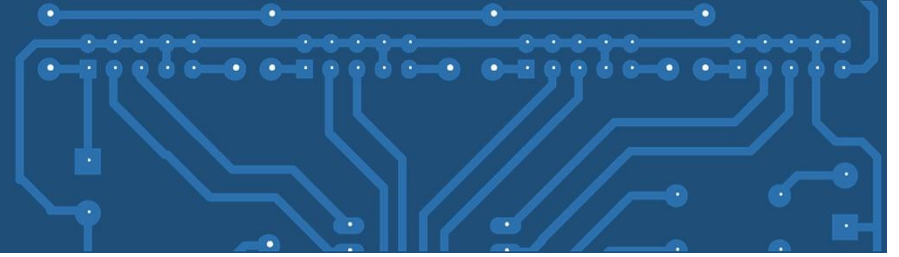
The wind frame $\mathcal{F}^w$

The origin of the body frame is identical to the vehicle frame, i.e. CoM of the AR.

$\mathcal{F}^w$ is obtained by rotating $\mathcal{F}^s$ in a **righthanded** rotation about the $\mathbf{k}^s$ axis by the **side-slip** angle β , such that the unit vector $\mathbf{i}^w$ aligns with the airspeed vector V_a .



$\mathbf{i}^w$ —axis points along the airspeed vector V_a .
 $\mathbf{j}^w$ —axis is constructed to make a right-handed coordinate system
 $\mathbf{k}^w$ —axis is identical to $\mathbf{k}^s$ —axis.



Transformations



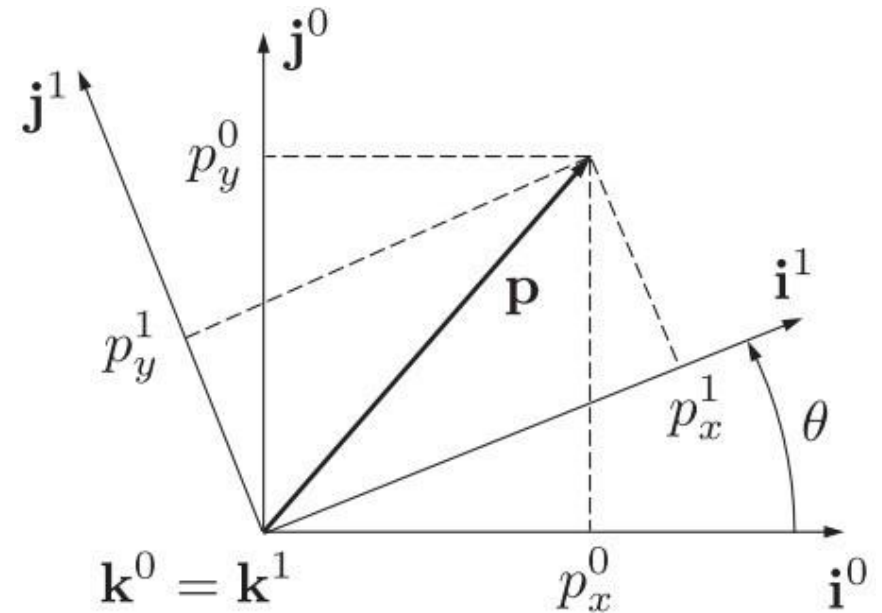
One vector, two reference frames

In Frame 0

$$\mathbf{p} = p_x^0 \mathbf{i}^0 + p_y^0 \mathbf{j}^0$$

In Frame 1

$$\mathbf{p} = p_x^1 \mathbf{i}^1 + p_y^1 \mathbf{j}^1$$



The tilted unit axes, expressed in Frame 0

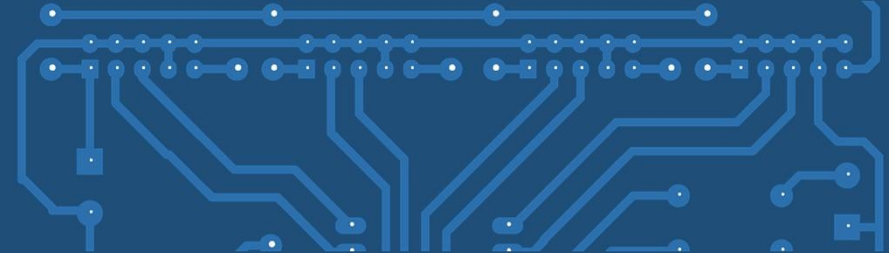
$$\mathbf{i}^1 = \cos\theta \mathbf{i}^0 + \sin\theta \mathbf{j}^0$$

$$\mathbf{j}^1 = -\sin\theta \mathbf{i}^0 + \cos\theta \mathbf{j}^0$$

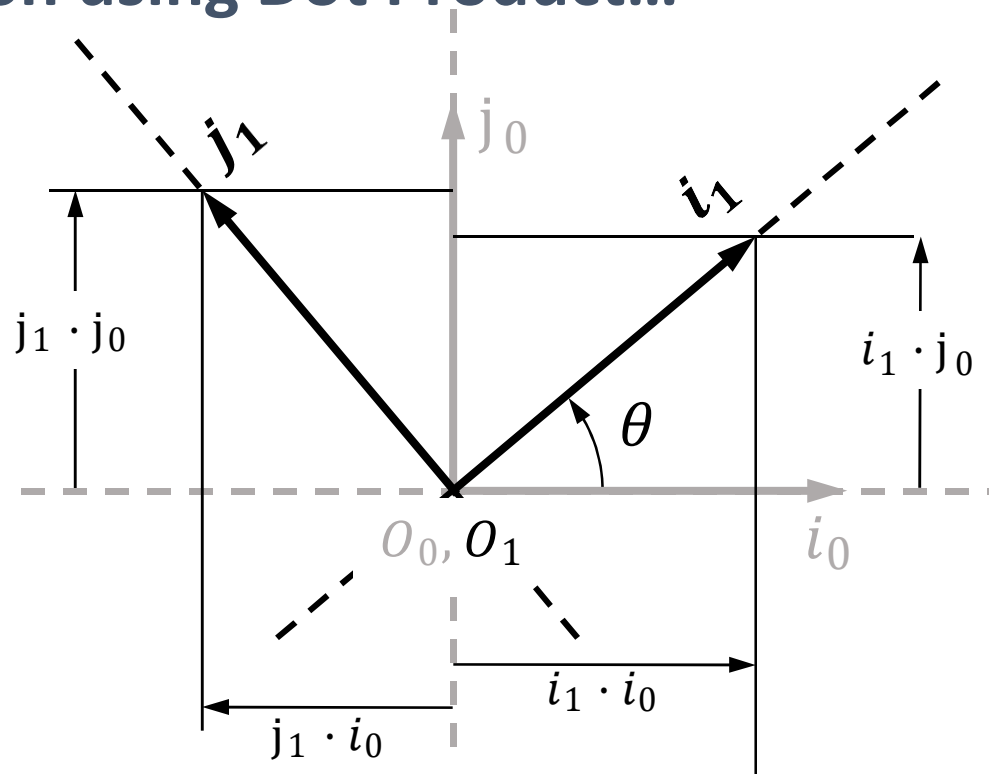
Convert coordinates from Frame 1 to Frame 0

$$\begin{bmatrix} p_x^0 \\ p_y^0 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} p_x^1 \\ p_y^1 \end{bmatrix}$$

Same vector. Different coordinates.



Rotation using Dot Product...



$$R_1^0 = \begin{bmatrix} i_1^0 & j_1^0 \end{bmatrix}$$

$$|i_1| = |j_1| = 1$$

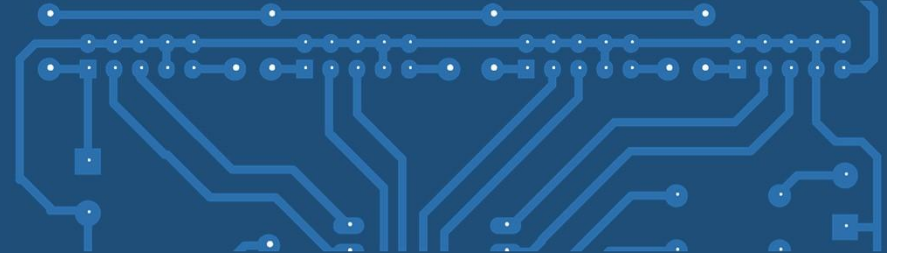
$$|i_0| = |j_0| = 1$$

$$i_1^0 = \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}$$

$$j_1^0 = \begin{bmatrix} -\sin \theta \\ \cos \theta \end{bmatrix}$$

$a \cdot b = |a| |b| \cos \theta$ = The dot product with a unit axis gives the signed component along that axis..

$$\therefore R_1^0 = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} = \begin{bmatrix} i_1 \cdot i_0 & j_1 \cdot i_0 \\ i_1 \cdot j_0 & j_1 \cdot j_0 \end{bmatrix}$$



Inverse rotation

Forward: convert Frame 1 coordinates into Frame 0

$$p^0 = R_1^0 p^1$$

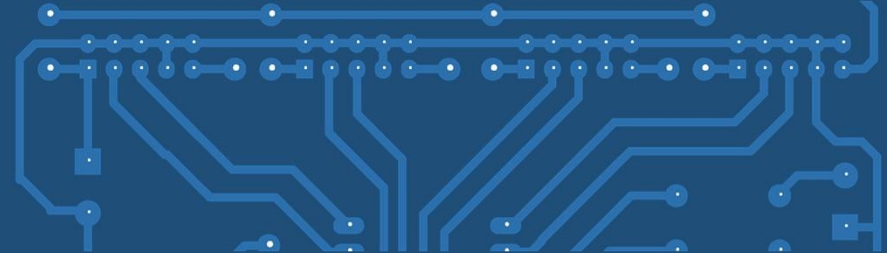
Reverse: convert Frame 0 coordinates into Frame 1

$$p^1 = R_0^1 p^0 \quad R_0^1 = (R_1^0)^{-1} = (R_1^0)^T$$

For a rotation matrix, transpose = inverse

$$R_1^0 = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \quad R_0^1 = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$$

Transpose: swap rows and columns. Equivalent to using $-\theta$.



Rotation-matrix properties

1. Reverse a transformation

$$R^{-1} = R^T \quad R^T R = I$$

Transforming there and back returns the original coordinates.

2. Combine successive transformations

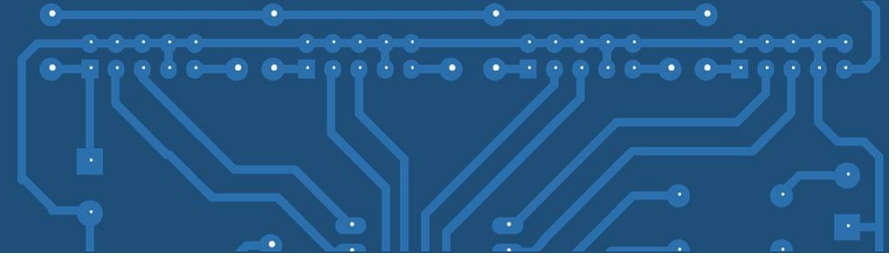
$$p^0 = R_1^0 R_2^1 p^2 \quad R_2^0 = R_1^0 R_2^1$$

Read right to left: Frame 2 to Frame 1, then Frame 1 to Frame 0.

3. Preserve geometry

$$\det(R) = +1$$

A proper rotation preserves lengths, angles and handedness.



Therefore, the rotation matrix R_0^1 that represents a right-handed rotation of angle θ about the z-axis.

$$R_0^1 = \begin{bmatrix} i^1 \cdot i^0 & i^1 \cdot j^0 & i^1 \cdot k^0 \\ j^1 \cdot i^0 & j^1 \cdot j^0 & j^1 \cdot k^0 \\ k^1 \cdot i^0 & k^1 \cdot j^0 & k^1 \cdot k^0 \end{bmatrix}$$

$$|i_0| = |j_0| = 1$$

$$|i_1| = |j_1| = 1$$

$$i_1 \cdot i_0 = \cos \theta$$

$$i_1 \cdot j_0 = \sin \theta$$

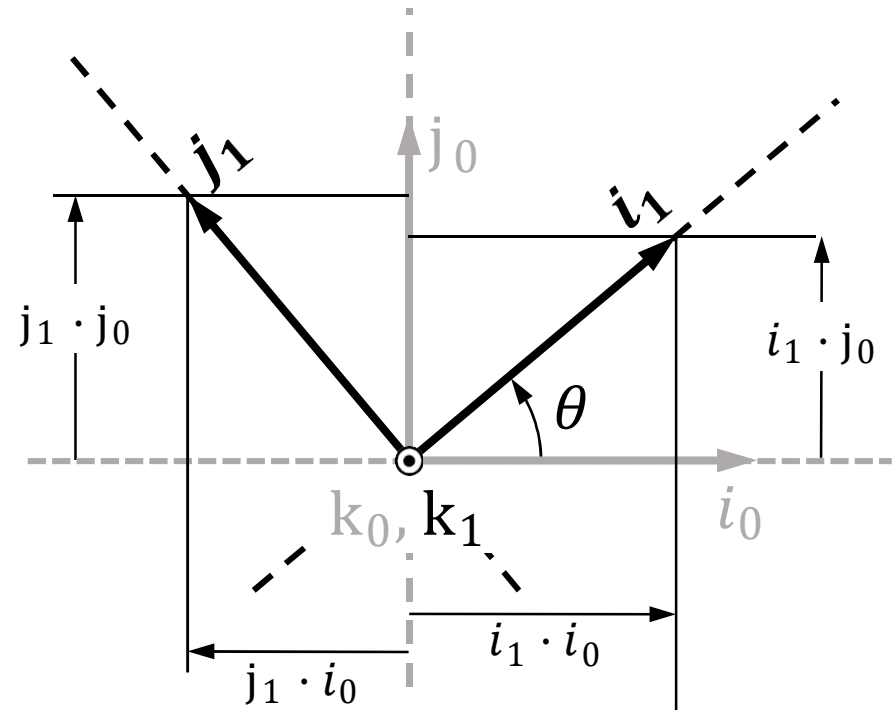
$$j_1 \cdot i_0 = -\sin \theta$$

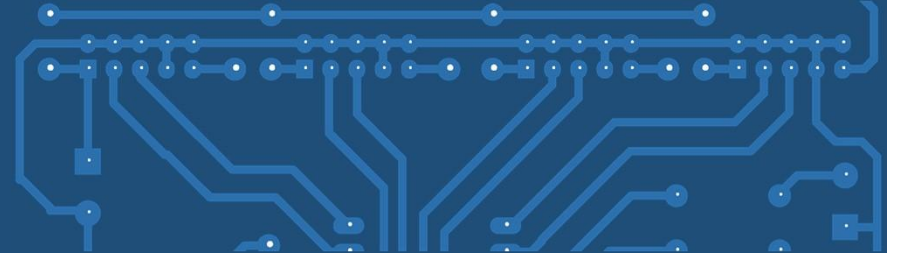
$$j_1 \cdot j_0 = \cos \theta$$

$$k_1 \cdot k_0 = 1$$

$$\text{The rest} = 0$$

$$\therefore R_0^1 = \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$





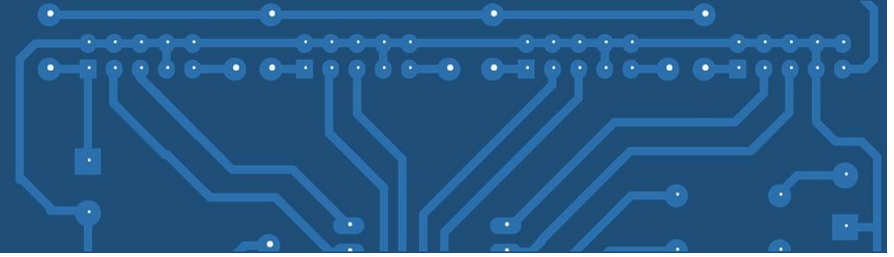
Similarly...

A right-handed rotation of angle θ about the y-axis.

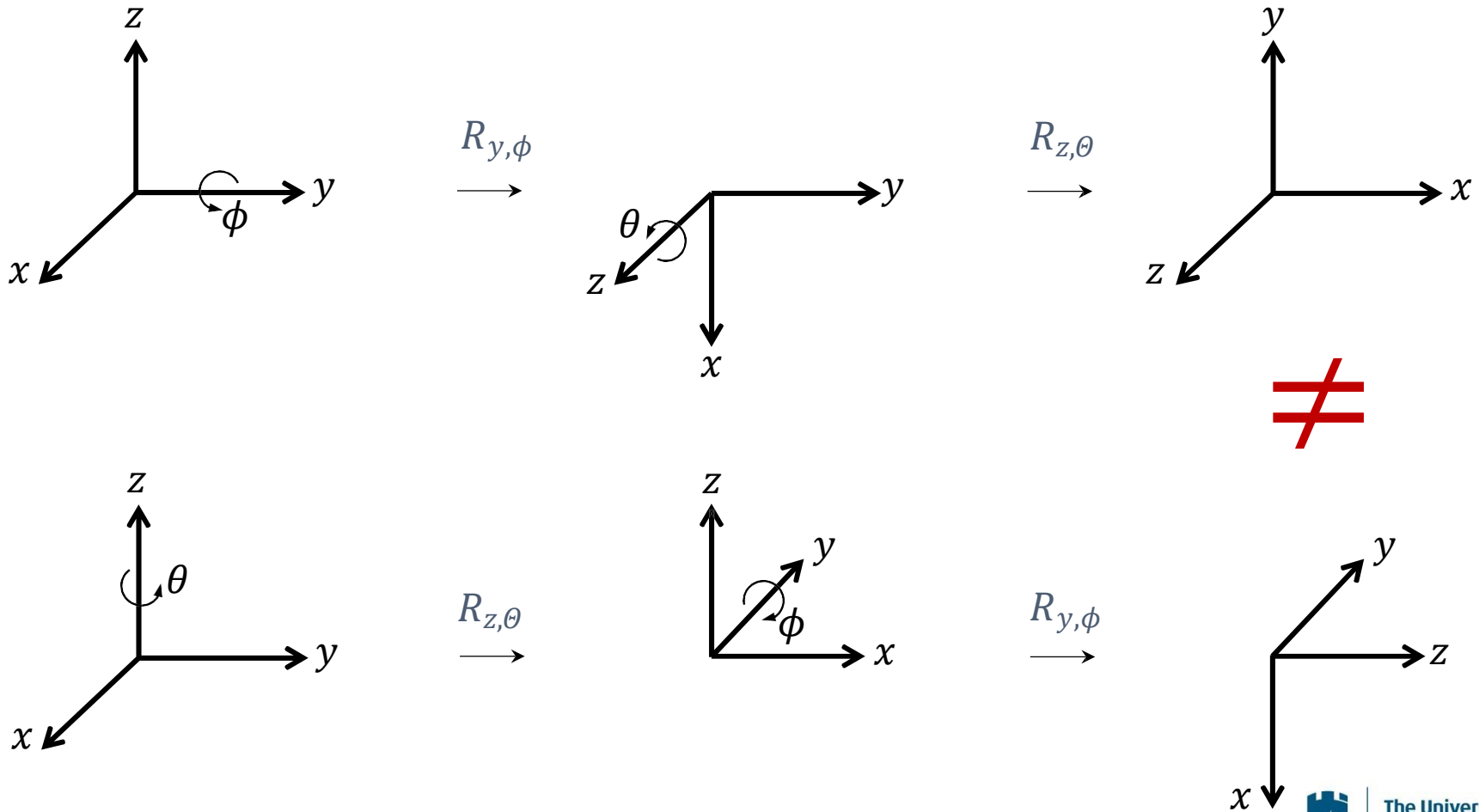
$$R_0^1 = R_{y,\theta} = \begin{bmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{bmatrix}$$

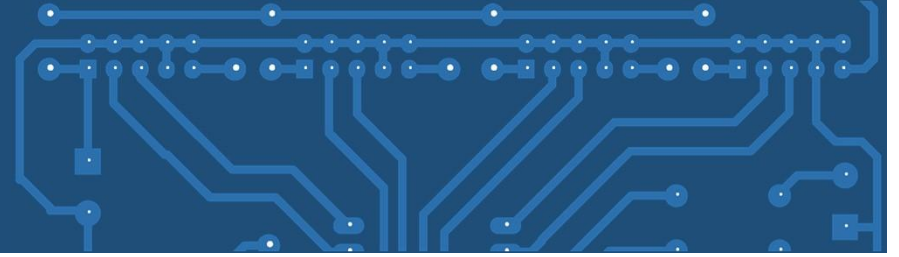
A right-handed rotation of angle θ about the x-axis.

$$R_0^1 = R_{x,\theta} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & -\sin \theta & \cos \theta \end{bmatrix}$$



Rotation order matters...

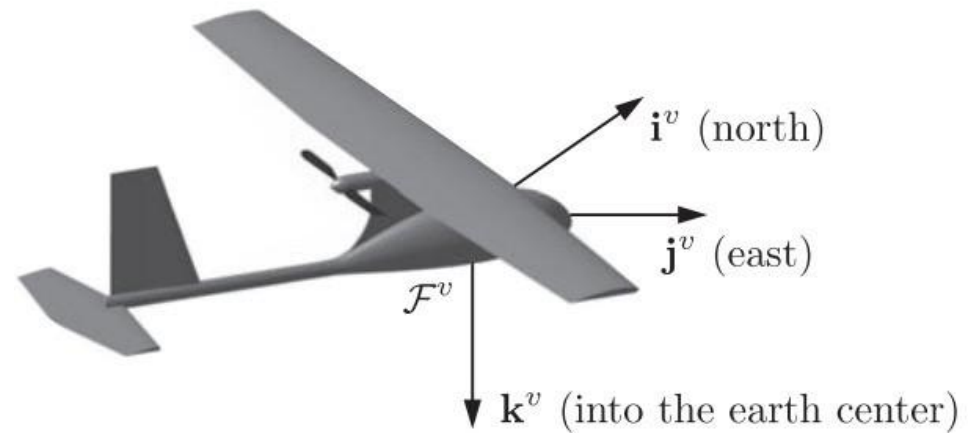
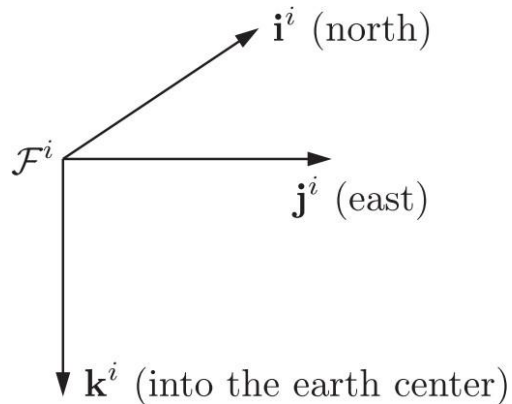




UAV Coordinates Frames Rotations



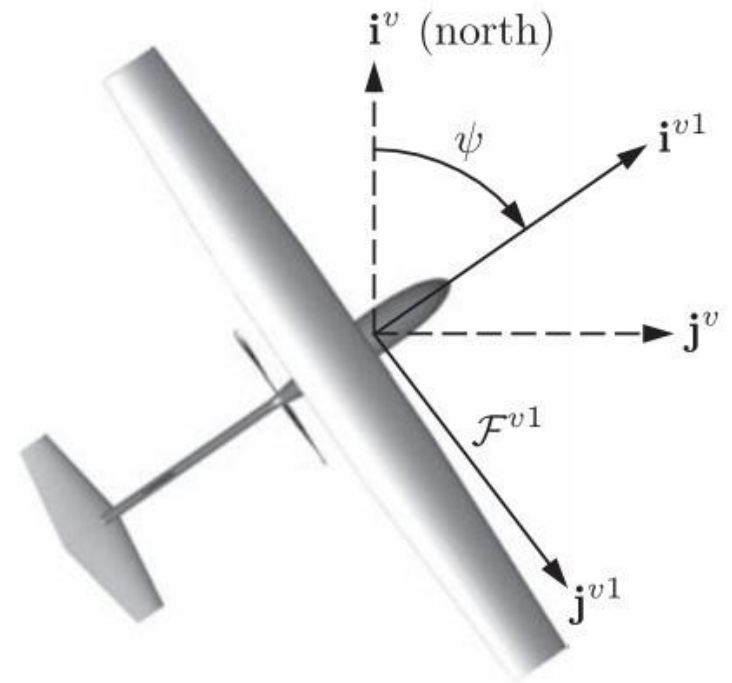
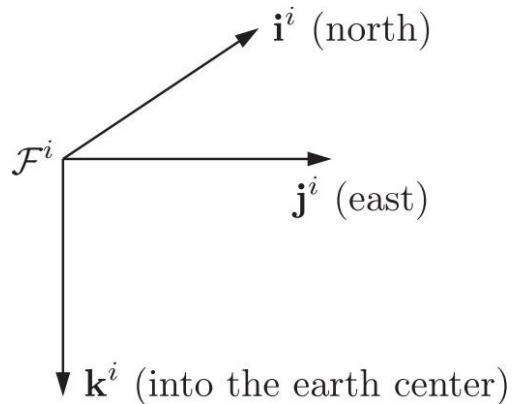
Transformation from inertial frame $\mathcal{F}^i$ to vehicle frame $\mathcal{F}^v$



Transformation from vehicle frame $\mathcal{F}^v$ to vehicle-1 frame $\mathcal{F}^{v1}$

$$\mathbf{p}^{v1} = \mathbf{R}_v^{v1} \mathbf{p}^v$$

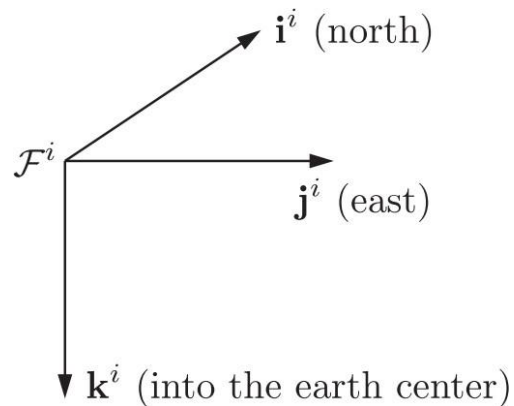
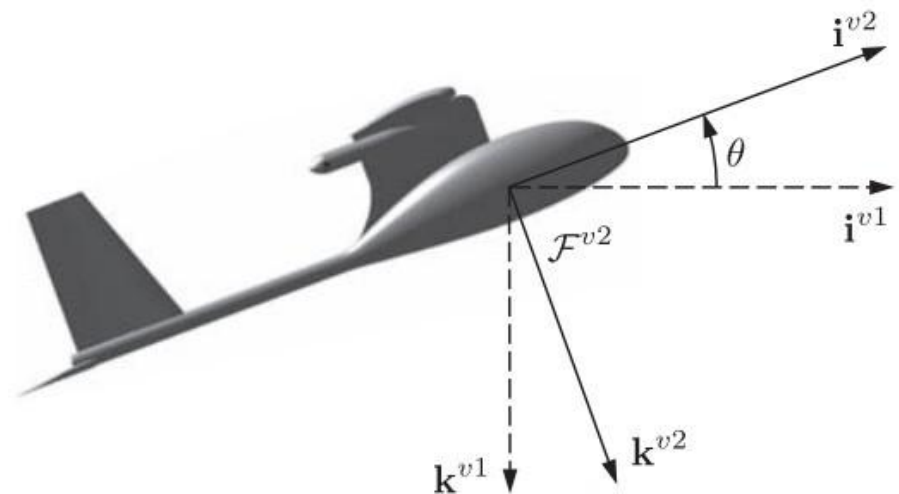
$$\text{where, } \mathbf{R}_v^{v1} = \begin{bmatrix} C\Psi & S\Psi & 0 \\ -S\Psi & C\Psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



Transformation from vehicle-1 frame $\mathcal{F}^{v1}$ to vehicle-2 frame $\mathcal{F}^{v2}$

$$\mathbf{p}^{v2} = \mathbf{R}_{v1}^{v2} \mathbf{p}^{v1}$$

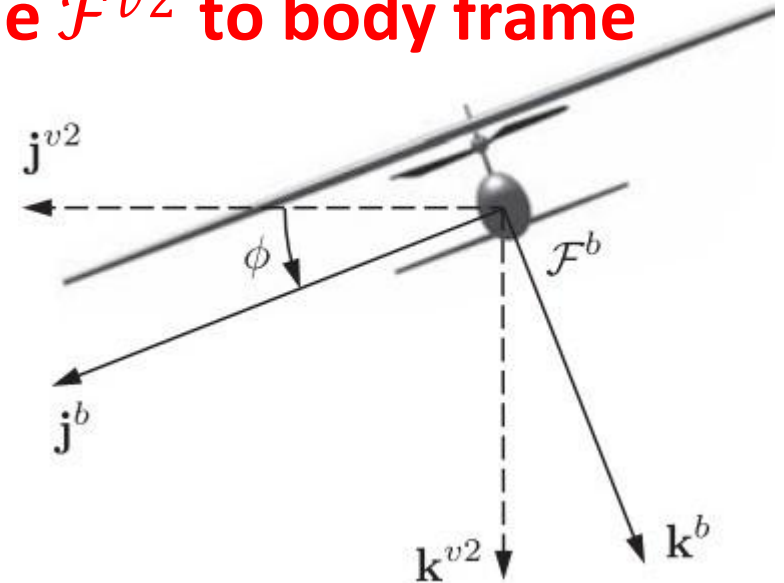
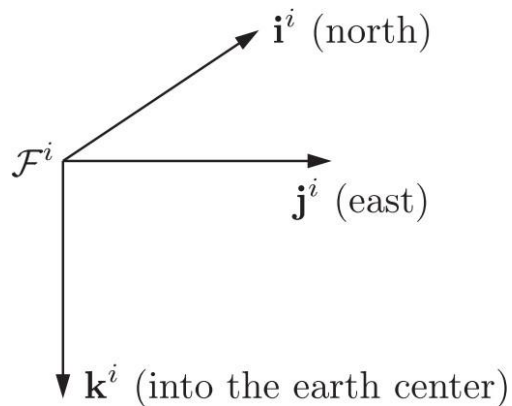
$$\text{where, } \mathbf{R}_{v1}^{v2} = \begin{bmatrix} C\theta & 0 & -S\theta \\ 0 & 1 & 0 \\ S\theta & 0 & C\theta \end{bmatrix}$$

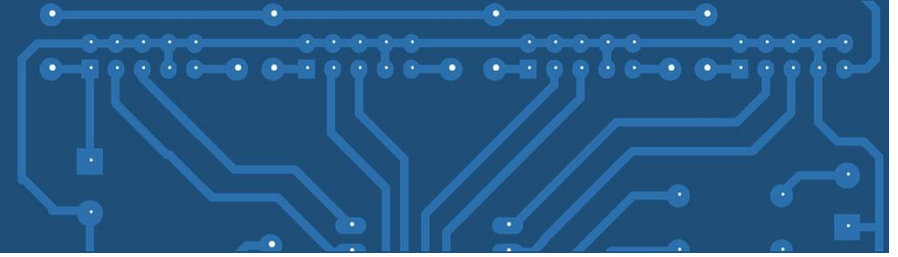


Transformation from vehicle-2 frame $\mathcal{F}^{v2}$ to body frame $\mathcal{F}^b$

$$\mathbf{p}^b = R_{v2}^b \mathbf{p}^{v2}$$

$$\text{where, } R_{v2}^b = \begin{bmatrix} 1 & 0 & 0 \\ 0 & C\phi & S\phi \\ 0 & -S\phi & C\phi \end{bmatrix}$$



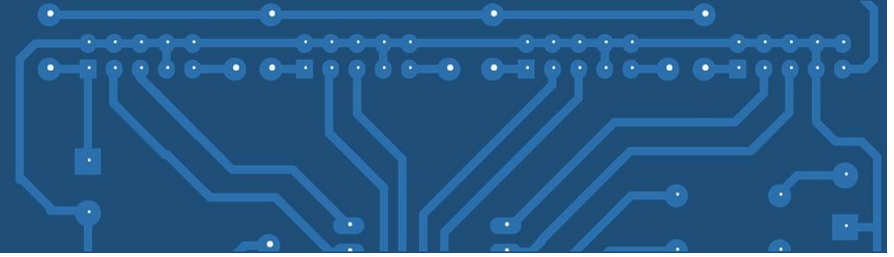


Transformation from vehicle frame $\mathcal{F}^v$ to body frame $\mathcal{F}^b$

$$\mathbf{p}^b = \mathbf{R}_v^b \mathbf{p}^v$$

where,

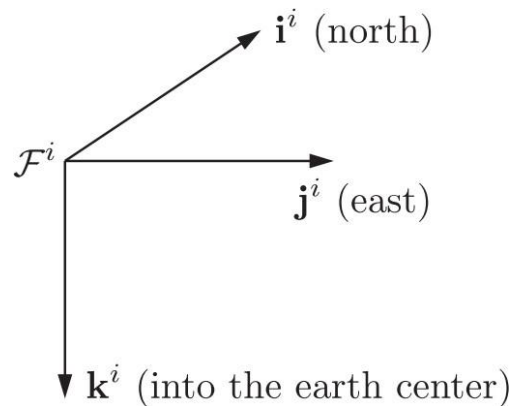
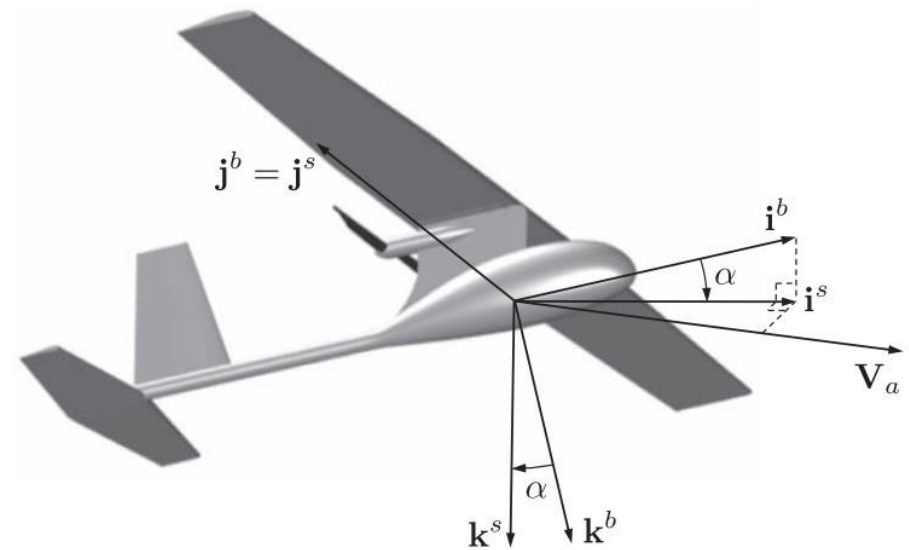
$$\begin{aligned} \mathbf{R}_v^b(\phi, \theta, \psi) &= \mathbf{R}_{v2}^b(\phi) \mathbf{R}_{v1}^{v2}(\theta) \mathbf{R}_v^{v1}(\psi) \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & C\phi & S\phi \\ 0 & -S\phi & C\phi \end{bmatrix} \begin{bmatrix} C\theta & 0 & -S\theta \\ 0 & 1 & 0 \\ S\theta & 0 & C\theta \end{bmatrix} \begin{bmatrix} C\psi & S\psi & 0 \\ -S\psi & C\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} C\theta C & C\theta S & -S\theta \\ S\phi S\theta C\psi - C\phi S\psi & S\phi S\theta S\psi + C\phi C\psi & S\phi C\theta \\ C\phi S\theta C\psi + S\phi S\psi & C\phi S\theta S\psi - S\phi C\psi & C\phi C\theta \end{bmatrix} \end{aligned}$$



Transformation from body frame $\mathcal{F}^b$ to stability frame $\mathcal{F}^s$

$$\mathbf{p}^s = R_b^s(\alpha) \mathbf{p}^b$$

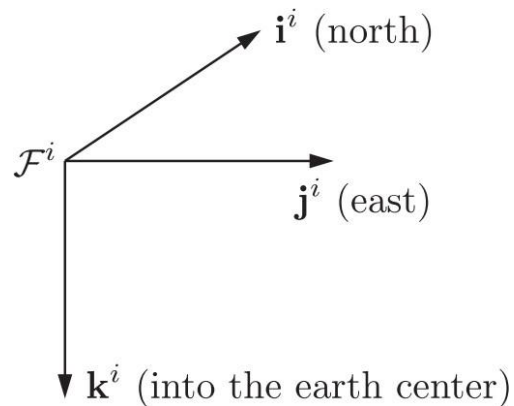
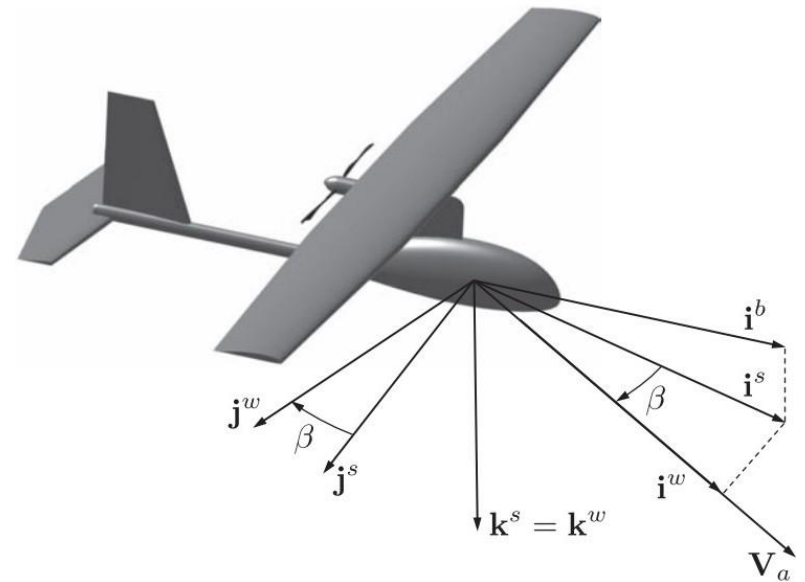
$$\text{where, } R_b^s = \begin{bmatrix} C\alpha & 0 & S\alpha \\ 0 & 1 & 0 \\ -S\alpha & 0 & C\alpha \end{bmatrix}$$

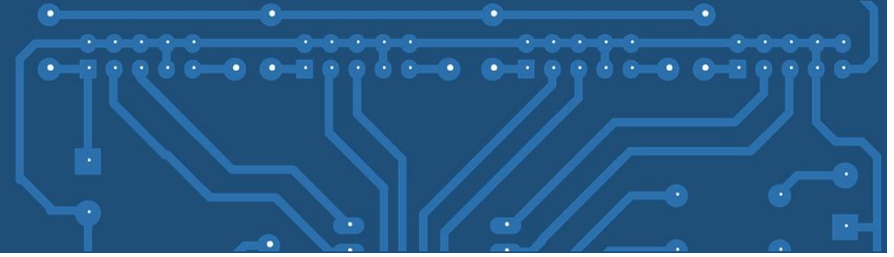


Transformation from stability frame $\mathcal{F}^s$ to wind frame $\mathcal{F}^w$

$$\mathbf{p}^w = R_s^w(\beta) \mathbf{p}^s$$

$$\text{where, } R_b^s(\beta) = \begin{bmatrix} C\beta & S\beta & 0 \\ -S\beta & C\beta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$





Transformation from body frame $\mathcal{F}^b$ to wind frame $\mathcal{F}^w$

$$\mathbf{p}^w = R_b^w(\alpha, \beta) \mathbf{p}^b$$

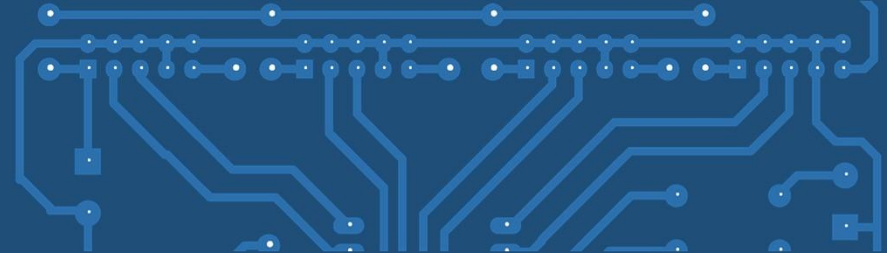
where,

$$\begin{aligned} R_b^w(\alpha, \beta) &= R_s^w(\beta) R_b^s(\alpha) \\ &= \begin{bmatrix} C\beta & S\beta & 0 \\ -S\beta & C\beta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} C\alpha & 0 & S\alpha \\ 0 & 1 & 0 \\ -S\alpha & 0 & C\alpha \end{bmatrix} = \begin{bmatrix} C\beta C\alpha & S\beta & C\beta S\alpha \\ -S\beta C\alpha & C\beta & -S\beta S\alpha \\ -S\alpha & 0 & C\alpha \end{bmatrix} \end{aligned}$$

Inverting the expression yields the inverse rotation (from the wind frame to the body frame):

$$R_w^b(\alpha, \beta) = (R_b^w)^T(\alpha, \beta) = \begin{bmatrix} C\beta C\alpha & -S\beta C\alpha & -S\alpha \\ S\beta & C\beta & 0 \\ C\beta S\alpha & -S\beta S\alpha & C\alpha \end{bmatrix}$$





Part 1 Quiz

Choose one answer for each question.

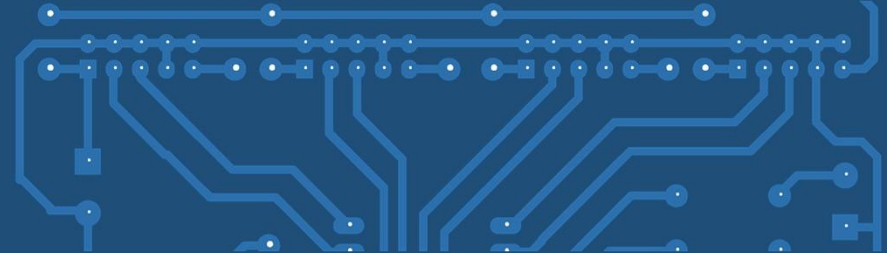
1. Which way does the positive body x-axis point?

Use the forward–right–down body-axis convention.

- A. North
- B. Forward, towards the aircraft's nose
- C. Vertically down towards Earth
- D. East

2. In $p^0 = R_1^0 p^1$, what does R_1^0 do?

- A. Squares the vector
- B. Converts Frame 0 coordinates into Frame 1
- C. Converts Frame 1 coordinates into Frame 0
- D. Moves the point to a new position



Part 1 Quiz

3. Frame 1 is turned 90° anticlockwise relative to Frame 0.

Both frames share an origin. Find p^0 when $p^1 = [2, 0]^T$.

$$p^0 = R_1^0 p^1 \quad R_1^0 = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

Use $\cos 90^\circ = 0$ and $\sin 90^\circ = 1$.

4. Which matrix reverses a rotation transformation R ?

A. $-R$ B. R^T C. $2R$ D. I

5. Does rotation order always give the same result?

Compare a turn about the current y-axis then the current z-axis with the same turns in reverse order. Explain briefly.